

Practice Quiz: Communication (Computational Neuroscience)

Name: __ Date: __

Part A: Multiple Choice

1. The temporoparietal junction (TPJ) is primarily involved in: A) Visual processing B) Theory of Mind — representing others' beliefs and intentions C) Motor control D) Auditory processing
2. The N400 ERP component occurs when: A) A word is grammatically incorrect B) A word is semantically unexpected in context C) A face is recognized D) A motor action is planned
3. The mirror neuron system supports social understanding by: A) Reflecting light from mirrors B) Internally simulating observed actions using motor regions C) Only processing verbal communication D) Storing memories of social interactions
4. Oxytocin modulates social cognition by: A) Decreasing attention to social cues B) Increasing the precision (gain) on social signals C) Blocking all social processing D) Only affecting romantic relationships
5. In the Active Inference account of autism, social difficulties may arise from: A) Lack of social motivation B) Under-weighting social priors and over-weighting sensory precision C) Damage to the language areas D) Excessive oxytocin
6. Broca's area primarily processes: A) Visual features B) Grammar (syntax) and speech production predictions C) Emotional prosody D) Face recognition
7. The amygdala's role in the social brain network is to: A) Process language semantics B) Evaluate social and emotional salience C) Control eye movements D) Store long-term memories

Part B: Short Answer

1. Explain how language processing works as predictive coding using the N400 and P600. Give an example sentence for each ERP and explain what is being predicted and what the prediction error represents.
2. Compare the mirror neuron system and mentalizing network. When would you rely primarily on the mirror system (fast, automatic) vs. the mentalizing network (slow, deliberate) for social understanding?
3. A researcher gives oxytocin to participants before a trust game and finds they become more trusting. Using the Active Inference precision framework, explain this finding. What are the potential benefits and risks of pharmacologically boosting social precision?